AI-POWERED PROMO FINDER

Kuku AI Generalist Assignment | Problem 1 - Find the Promos (Ads)

A reproducible editorial decision-support tool that extracts and ranks 20-60 second promotional clips from timestamped microdrama screenplays. The tool shortlists; a human editor makes the posting decision.

1. Problem choice, framing and assumptions

Why this problem: Editors manually review every new episode to find compelling ad moments. This is recurring, time-intensive and directly tied to viewer acquisition. I chose a narrow, verifiable intervention: suggest usable clip boundaries and reasons, not edit videos or predict real campaign performance.

Assumptions: Scripts approximate on-screen timing; eligible clips last 20-60 s, hook within 3 s, and should end with a meaningful unresolved question. Continuous clips are prioritized; the architecture supports ordered segments but cross-episode discovery is not yet implemented. The tool only reads screenplay text; it cannot judge acting, actual pacing or visual execution.

2. Working approach - code for precision, AI for narrative judgment

Stage	Method and control
Parse	Python extracts episode/scene and overlapping CAMERA, ACTION, SOUND FX and DIALOGUE beats, with stable IDs (e.g., E01-B0042).
Discover	Gemini analyzes episode beats and selects hook/end beat IDs and narrative rationale; it does not fabricate timestamps.
Validate	Code resolves IDs to original timestamps, checks 20-60 s duration, hook within 3 s, ending evidence, ordering, quote support and legal boundaries; rejects/retries invalid picks.
Judge + rank	Separate blind LLM judgment sees clip text without logline; scores hook 30%, cliffhanger 30%, standalone 25%, emotion 15%. Code caps generic cliffhanger questions, checks evidence and limits overlap/episode concentration.

Important timing detail: Camera, action and dialogue often overlap, so continuous duration is clip end minus clip start, never the sum of individual beat durations. Invalid candidates and raw LLM responses are retained for audit.

3. Full-pack run: measured results

Input and output	Reliability / computation
50/50 episodes processed; 58 validated and ranked promos; 37 rejected candidate records	0 judge failures; 0 duplicates removed; 12/12 automated tests passing
Top-10 mean LLM score 7.63/10 vs random baseline 5.90/10	125 API calls; 7 cache hits; 590.8 s total (~9 min 51 s)

These are one full-pack run results, not estimates of marketing lift. The 7.63 vs 5.90 comparison uses the same automated judge and therefore is not an independent efficacy measure.

4. Highest-ranked recommendations (full-pack output)

Rank	Episode / cut points	Duration	LLM score	Moment / ending
1	E18 00:00.00-00:48.80	48.8 s	8.00	Fire rescue; escape still in progress
2	E23 01:18.50-02:11.50	53.0 s	7.90	Poisoning revelation; door opens during antidote
3	E36 02:36.50-03:20.75	44.2 s	7.75	Exiled Duke vows revenge
4	E13 00:53.50-01:43.00	49.5 s	7.70	Bloodied search for ring remains unresolved
5	E24 00:30.80-01:29.80	59.0 s	7.70	Public whipping; defiant vow
6	E21 00:03.60-00:35.80	32.2 s	7.70	Urgent plea to save a dying child
7	E34 00:00.00-00:54.80	54.8 s	7.55	Witchcraft trial; maid answers accusation
8	E43 00:57.00-01:52.80	55.8 s	7.50	Conspiracy exposed; fugitive blocked by police
9	E07 01:05.20-01:28.50	23.3 s	7.30	Bully confronted by intimidating husband
10	E33 01:04.20-01:59.50	55.3 s	7.25	Maid coerced into false testimony

QUALITY, LIMITATIONS AND DELIVERY

Evaluation criteria are deliberately strict: a dramatic payoff is not necessarily a cliffhanger, and a valid timestamp is not proof of an effective ad.

5. Quality checking - transcript-based editorial assessment

I evaluated 20 screenplay clip intervals (10 AI shortlist selections and 10 randomized comparison clips) against four binary quality criteria and a posting recommendation. This is **AI-assisted transcript-based editorial assessment**, not independent human viewing or observed audience behavior.

Criterion (pass count / 10)	AI shortlist	Random baseline	Interpretation
Hook stops scroll in first 3 s	9 / 10 (90%)	1 / 10 (10%)	Large advantage in immediate openings
Ending genuinely unresolved	6 / 10 (60%)	6 / 10 (60%)	No observed cliffhanger advantage
Clear without backstory	9 / 10 (90%)	8 / 10 (80%)	Small advantage, limited sample
Timestamps correct	10 / 10 (100%)	10 / 10 (100%)	Boundary checks work on assessed clips
Would recommend posting	6 / 10 (60%)	0 / 10 (0%)	Editorial proxy, not measured ad performance

Interpretation: Selection substantially improved hook quality in this small assessed set, but did not outperform random clips on unresolved endings. Twenty transcript clips are too few to establish production performance; the same judge drives shortlist ranking, so the manual-style transcript check is a useful second lens, not a replacement for editor/video review or live A/B tests.

6. Specific failures and what changed

Observed failure	Evidence and response
Invalid clip length	E03 initially proposed an 87.5 s stranger-rescue clip; code rejected it. Precise retry feedback recovered a valid 54.2 s clip. The validator did not silently trim a bad pick.
Resolved ending mislabeled a cliffhanger	E34 ends after the maid says “Yes”; the supposed uncertainty about whether she would testify was already answered. E24 closes on a completed defiant vow; E36 closes on a revenge declaration. These should be down-ranked or rejected by an editor.
Low standalone clarity / payoff	E43 opens with a dossier reference that assumes context, then a fleeing suspect is blocked by police - a capture payoff rather than a clean unresolved cut.
Judge variability and malformed output	Prompt changes moved individual cliffhanger ratings by as much as 3 points on the same clip. Earlier missing schema fields created artificial 0.0 scores; required-field retries and explicit judge-failed handling fixed this (0 failures in final run).

7. Prototype, reproducibility and scale

Prototype: Python CLI (`python run.py --input pack.zip --out outputs/final`) and Streamlit UI for ZIP upload, model selection, ranked clips and JSON/CSV export. Each output includes episode, ordered exact cut points, hook, cliffhanger, reasons, scored evidence and warnings. Raw LLM responses, validated/rejected candidates, rankings, summary and review files are saved for inspection.

Scale and cost: Episodes are independently processed and cacheable; retry/backoff protects transient API failures. A single provider abstraction supports Gemini, Groq, Anthropic, OpenAI-compatible endpoints and a heuristic offline baseline. On this 50-episode run: 125 calls and 590.8 s. Provider quotas, pricing, video review and editorial labor still govern real deployment; no guaranteed free-tier capacity or per-show dollar cost is claimed.

Human-in-loop: An editor checks transcript/footage alignment, cut naturalness, standalone appeal, context and brand fit before publishing. After launch, compare hook retention, watch-through and click-through in randomized creative tests. Confidential Kuku screenplays, caches and derived outputs must not be published in a public repository.

8. AI usage and next iteration

Tools: Gemini API for candidate discovery and blind narrative judgment; Claude and ChatGPT for planning, debugging, rubric/evaluation critique and documentation support; Python/Pydantic/Pandas for deterministic processing and validation; Streamlit for interface. **Where AI got it wrong:** Flash-Lite repeated the same invalid 87.5 s selection on first retry; judge prompts also mistook resolved reactions or revenge vows for cliffhangers. Explicit validator feedback, score caps and schema checks reduced these errors, but did not eliminate cliffhanger ambiguity.

Next: Improve end-of-clip question-resolution checking; evaluate with independent human editors viewing actual footage; measure real campaign impact. Source code, private raw outputs and this PDF are separate submission artifacts.